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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **ITA1439**

Course Name: **Ethical Hacking**

Duration: **1: 30 Hr**

Date: **02-11-2026**

36/50 72%

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes
On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply the principle of the foundational concepts of ethical hacking, including types of ha principles, legal frameworks, and the role of responsible disclosure in cybersecurity.
CO2	Execute footprinting in social engineering such as DNS analysis, competitive intel gathering, and attack simulations to collect relevant information for identifying pote risks.

Answer all the Questions

1. A private bank hires a security firm to test its internet-banking portal before la month, a former employee uses old credentials to view customer records, a student attack tool against the portal "to see what happens", and an activist group defaces Define the term hacker and classify each person or group, including the hired t black hat, grey hat, script kiddie, hacktivist or insider, with reasons. Explain the e testers must follow, such as written authorization, agreed scope, confidentiality, honest reporting.

2. A student finds an SQL injection flaw in the e-learning portal of another univers marks and contact details of about 20,000 students, without having permission to it. The student is considering posting the details on social media, reporting privat or offering the details in exchange for payment. Analyze the legal position of

Cyber Security
Report

Cyber Security

Ethical Hacker

An Ethical Hacker is a computer and network expert who possesses technical skills of Computer System, networks, applying security mechanisms and uses that to understand and analyse.

They are classified according to their authorization and the nature of activities.

Ethical Hacker

An Ethical Hacker is an ethical Security threat who performs authorized security vulnerability.

Activities are legal and ethical when done within the authorized scope.

Bad Hacker

A bad hacker accesses computer systems

QL Injection

QL Injection is a web application vulnerability which an attacker's manipulates SQL queries access or modify database info.

Legal position under the IT Act, 2000

The Info Tech Act 2000, provides legal provisions address unauthorized process and misuse of computer system's.

Section 43 - Unauthorized Access:

It cover's accessing a computer system without permission.

- Downloading or copying info
- Extracting data from a computer system.
- Damaging computer systems.

Section 66 - Computer Related Offences.

Section 66 applies when an act under section 43 is committed dishonestly or fraudulently

It may result in fine upto 5 lakhs

→ The student intention and actions determine wether criminal liability applies.

Transfer.

- any DNS server stores DNS records
- The Secondary server requests a zone transfer
- The primary server checks authorization

Conclusion

DNS zone transfer helps maintain synchronization b/w primary and secondary DNS server
unauthorised transfers.

...engineering attacks.

...ial engineering is a technique in which ...ker manipulate people into revealing ...fidential information.

...dentification of Attacks.

phishing

...shing is an attack in which fake emails ...e sent to employes to steal passwords personal info. In this scenario, the attacker ...ends a fake IT help-desk.

vishing

...ishing is a voice-based phishing attack ...erformed through telephone calls.

• pretexting :

...retexting involves creating a fake identity ...or believable story to obtain confidential information. For example.

- tailgating

tailgating occurs when an unauthorized person follows an authorized employee.

conclusion

Social engineering attacks exploit human. behav...

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